STA 5635 HW 4

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Problem 1

TISP variable selection method ith threshold operator:

$$\Theta(x, \lambda, \eta) = \begin{cases} 0 & \text{if} |x| \le \lambda \\ \frac{x}{1+\eta} & |x| > \lambda \end{cases}$$

Penalized logistic regression:

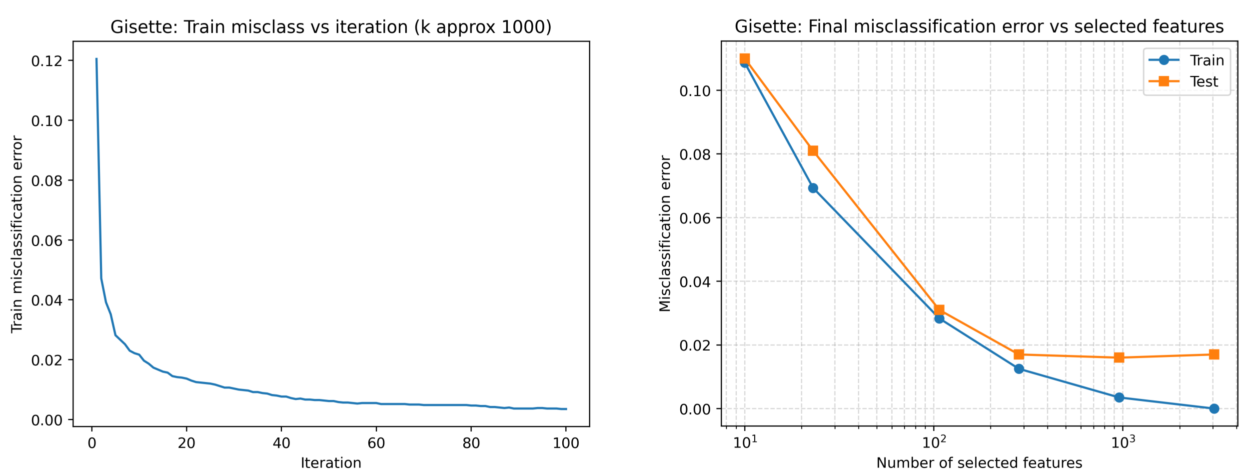
$$L(\boldsymbol{\omega}) = \frac{1}{N} \sum_{j=1}^{N} \ln(1 + e^{-y_j^* \boldsymbol{\omega}^T \boldsymbol{x}_j}) + \lambda \sum_{i=1}^{p} p(\omega_i)$$

assuming $y_j^* \in \{-1, 1\}$ and algorithm iteration:

$$\boldsymbol{\omega}^{t+1} = \Theta\left(\boldsymbol{\omega}^t + \eta' X^T \left[\boldsymbol{y} - \frac{1}{1 + e^{-X\boldsymbol{\omega}^t}}\right], \lambda\right)$$

assuming $y_j \in \{-0, 1\}$.

First portion of question 1 is shown in Figure 1:



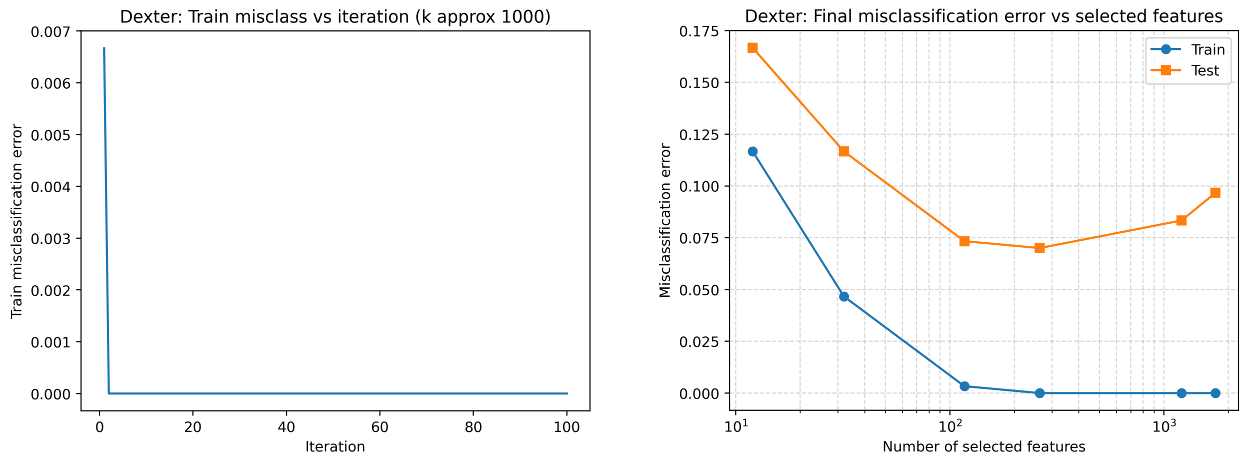
(a) Training misclassification error vs Iteration Number

(b) Train and Test Misclassification error vs Number of Selected Features

Figure 1: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Gisette Dataset.

Second portion of question 1 is hown in Figure 2

The table reporting the train and test misclassification error, the test AUC, corresponding number of features selected, values of λ, and the training time are reported in



(a) Training misclassification error vs Iteration Number

(b) Train and Test Misclassification error vs Number of Selected Features

Figure 2: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Dexter Dataset.

Problem 2

FSA variable selection using logistic loss with shrinkage:

$$L_D(\beta) = \frac{1}{N}\sum_{i=1}^{N} \ln(1 + e^{-y_i \beta^T \boldsymbol{x}_j}) + s\|\beta\|_2^2$$

Inverse scheduler with parameter μ:

$$M_i = k + (p-k)\max\left(0, \frac{N^{iter} - 2i}{2i\mu + N^{iter}}\right)$$

dataset	target$_{f}eatures$	selected$_{f}eatures$	selected$_{l}ambda$	train$_{m}isclass$	test$_{m}isclass$	test$_{a}uc$	t
Gisette	10	10	0.196304	0.108667	0.110000	0.942920	
Gisette	30	23	0.138489	0.069333	0.081000	0.972464	
Gisette	100	107	0.086975	0.028333	0.031000	0.993964	
Gisette	300	282	0.054623	0.012500	0.017000	0.997420	
Gisette	1000	956	0.019179	0.003500	0.016000	0.998280	
Gisette	3000	3048	0.004229	0.000000	0.017000	0.997964	
Dexter	10	12	0.138489	0.116667	0.166667	0.932844	
Dexter	30	32	0.097701	0.046667	0.116667	0.959911	
Dexter	100	117	0.068926	0.003333	0.073333	0.975733	
Dexter	300	263	0.054623	0.000000	0.070000	0.979600	
Dexter	1000	1210	0.038535	0.000000	0.083333	0.968400	
Dexter	3000	1739	0.030539	0.000000	0.096667	0.966978	

Table 1: Train and Test Misclassification Errors, Test AUC, Number of Selected Features, values of λ, and training time.